

## Activity 3 : Technology Comparison Matrix

### 3.1 Consolidated Candidate Stacks

Three complete technology stack options were consolidated from the frontend, backend, database, and authentication comparisons in Activities 1 and 2, and scored against a common set of weighted criteria reflecting FitFlow's priorities: performance, scalability, development speed, security/compliance, cost, AI/ML support, real time capability, and maintainability.

- **Option A** : React Native (+RN Web) • Node.js/Express • PostgreSQL + Firebase • Firebase Auth • TensorFlow Lite + Cloud AI
- **Option B** : Flutter • Python/FastAPI • MongoDB • Auth0 • Cloud AI
- **Option C** : Kotlin Multiplatform • Go • DynamoDB • AWS Cognito • Cloud AI

### 3.2 Weighted Decision Matrix

Each criterion was assigned a weight (summing to 100%) reflecting its importance to the FitFlow redesign, and each option was scored from 1 (poor) to 5 (excellent). The weighted score is the sum of (criterion weight × raw score).

| Criteria              | Weight | Option A (Score / Wtd) | Option B (Score / Wtd) | Option C (Score / Wtd) |
|-----------------------|--------|------------------------|------------------------|------------------------|
| Performance           | 15%    | 4 / 0.60               | 5 / 0.75               | 5 / 0.75               |
| Scalability           | 12%    | 4 / 0.48               | 4 / 0.48               | 5 / 0.60               |
| Development Speed     | 15%    | 5 / 0.75               | 3 / 0.45               | 2 / 0.30               |
| Security & Compliance | 15%    | 4 / 0.60               | 4 / 0.60               | 4 / 0.60               |
| Cost                  | 10%    | 5 / 0.50               | 3 / 0.30               | 3 / 0.30               |
| AI / ML Support       | 15%    | 4 / 0.60               | 5 / 0.75               | 2 / 0.30               |
| Real time Capability  | 10%    | 5 / 0.50               | 3 / 0.30               | 3 / 0.30               |
| Maintainability       | 8%     | 5 / 0.40               | 3 / 0.24               | 2 / 0.16               |
| Weighted Total        | 100%   | 4.43                   | 3.87                   | 3.31                   |

Table 3.1: Weighted decision matrix across the three consolidated technology stacks

### 3.3 Recommended Technology Stack

Option A React Native (+ RN Web) • Node.js/Express • PostgreSQL + Firebase • Firebase Auth • TensorFlow Lite with a Cloud AI service : achieves the highest weighted score (4.43 / 5) and is the recommended stack for the FitFlow redesign.

- **Rationale:** It ties for the lead on development speed and maintainability, both critical given FitFlow's mid sized team and urgency to reverse declining retention.
- Its lower per criterion raw scores on performance and AI/ML support versus Options B and C are outweighed by decisively higher scores on cost, real-time capability, and maintainability the criteria most exposed to FitFlow's team size and timeline constraints.
- Option B (Flutter/FastAPI/MongoDB/Auth0) scores well on raw AI/ML and performance but is penalised for splitting the team across two languages and lacking a native real time data layer.
- Option C (Kotlin Multiplatform/Go/DynamoDB/Cognito) has the best raw performance and scalability but is penalised heavily for slow development speed, weak AI/ML ecosystem fit, and the highest long term maintenance burden (three effectively separate codebases).